HUNNY TUTORIALS
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Class X Maths Test

Ch- Area Related to Circles

Date: 31.10.2026 **Time : 60 Min** **Marks: 30**

Section A (1 Marker Each)

Q.1. If the area of a circle is 154 cm^2 , then its perimeter is
(a) 11 cm (b) 22 cm (c) 44 cm (d) 55 cm

Q.2. The diameter of a car wheel is 21 cm. The number of complete revolutions it will make in moving 66 km is:
(A) 10^4 (B) 10^5 (C) 10^6 (D) 10^7

Q.3. The arc of a circle is of length 6π cm and the sector it bound has an area of 24π cm^2. The radius of the circle is:
(a) 4 cm (B) 8 cm (C) 16 cm (D) 18 cm

Q.4. If the perimeter of a circle is equal to that of a square, then the ratio of their areas is:
(a) 22:7 (b) 14:11 (c) 7:22 (d) 11:14

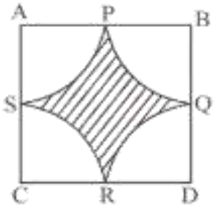
Q.5. It is proposed to build a single circular park equal in area to the sum of areas of two circular parks of diameters 16 m and 12 m in a locality. The radius of the new park would be
(a) 10 m (b) 15 m (c) 20 m (d) 24 m

Section B (2 Marker Each)

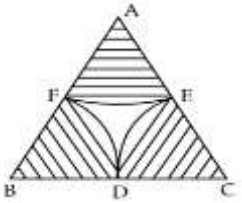
Q.6. Find the area of a sector of a circle with diameter 28 cm, If the length of the corresponding arc is 22 cm.

Q.7. Find the area of the minor segment of a circle of radius 14cm, when the angle of the corresponding sector is 60°.

Q.8. Find the area of the shaped region in Fig., where arcs drawn with centres A, B, C and D intersect in pairs at mid-points P, Q, R and S of the sides AB, BC, CD and DA, respectively of a square ABCD (Use π = 3.14).

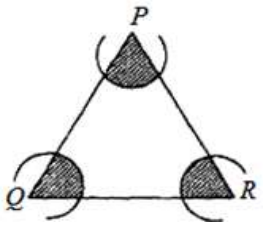


Q.9. In Fig., arcs are drawn by taking vertices A, B and C of an equilateral triangle of side 10cm. to intersect the sides BC, CA and AB at their respective mid-points D, E and F. Find the area of the shaded regions (Use π = 3.14).



OR

In Fig., arcs have been drawn with radii 14cm each and with centres P, Q and R. Find the area of the shaded region.



Section C (3 Marker Each)

Q.10. With the vertices A, B and C of a triangle ABC as centres, arcs are drawn with radii 5cm each as shown in Fig., If AB = 14cm, BC = 48cm and CA = 50cm, then find the area of the shaded region. (Use $\pi = 3.14$).



OR

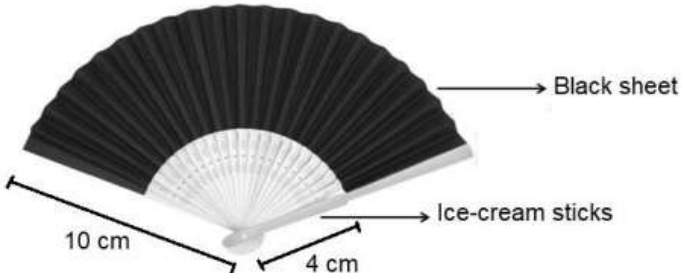
Sides of a triangular field are 15m, 16m and 17m. With the three corners of the field a cow, a buffalo and a horse are tied separately with ropes of length 7m each to graze in the field. Find the area of the field which cannot be grazed by the three animals.

Q.11. The length of the minute hand of a clock is 5cm. Find the area swept by the minute hand during the time period 6:05 am and 6: 40 am.

Q.12. Find the number of revolutions made by a circular wheel of area $1.54m^2$ in rolling a distance of 176m.

Section E (4 Marker Each)

Q.13. An Art and craft teacher prepared a fan which looked like the sector of a circle using ice-cream sticks and black sheet as shown in the given figure:



(i) The angle subtended at the centre of the sector is 60º. Find the area of one face of the fan. (1)

(ii) With the given sector angle of 60º at the centre, find area of black sheet used in the fan. (1)

(iii) (A) The boundary of the black sheet region on one face is to be covered by a designer tape. What length of tape is required? (2)

OR

(iii) (B) If 29.6 cm of designer tape is required to cover the boundary of the black sheet region on one face of the fan, then evaluate the central angle of the sector.

Q.14. A motion sensor (or motion detector) is an electronic device that is designed to detect and measure movement. The motion detector can detect movement over a sector of angle $\theta = 70°$ to a distance of 24m.

(i) Derive a relation between length of arc(l) and the area of the sector(A) enclosed by it? (1)

(ii) How much area is monitored by motion detector? (1)

(iii) (A) To increase the area to be monitored by 50% with same range of distance covered, what should be the angle? (2)

OR

(iii) (B) For $\theta = 91°$, what range of distance is required for the detector to monitor 30% more area?